

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		A: guard (cell) and B: epidermal (cell) Reject epithelium/ endodermal	1			1		1
		(ii)		Any four (×1) from A. K ⁺ actively transported into {cell A/ guard cell} (1) B. starch converted to malate (1) C. Lowers the water potential of the cell (1) D. Water moves into the cell by <u>osmosis</u> (1) E. (due to increase water) Cells {swell up / become turgid} (1) F. (Stomata opens): as the inner cell wall is {thicker / less elastic} than the outer cell walls / walls are unevenly thickened (1) If candidates refer to stomata rather than cells award maximum of 3 marks	4			4		
		(iii)		reduce water loss (1) reject prevent	1			1		
	(b)	(i)		Any two for one mark Temperature Humidity Light <u>intensity</u> / duration of light Volume of water / frequency of watering <u>Soil</u> pH Concentration of {Nutrients/ named nutrient}			1	1		1
		(ii)		As CO ₂ concentration increases {stomatal density decreases/ fewer stomata} / ORA(1)			1	1		1

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		(iii)		Any three (x1) from A. Stomata <u>open</u> to allow CO ₂ (to diffuse) in (1) B. for photosynthesis (1) C. (At higher CO ₂ concentrations) there is {an increased rate/ more} of CO ₂ {uptake/ diffusion} / ORA (1) D. Sufficient CO ₂ can be absorbed with fewer stomata/ ORA (1) E. Therefore less water lost from the plant (1)		2	1	3		
	(c)			- CO₂ concentrations were decreasing (in the Mesozoic era)/ younger rocks have a lower CO₂ concentration/ older rocks have a higher CO₂ concentration (1) - CO₂ concentrations were {<u>higher</u> (in the Mesozoic era than the present day)/ greater than 560 ppm} / ORA (1)			2	2		2
				Question 3 total	6	2	5	13	0	5